Risk & Return

Professor Philip Howard
howardpd@wfu.edu

WAKE FOREST
UNIVERSITY
SCHOOL of BUSINESS

Returns

- Return R_{t+1} at time $t+1$ is defined

$$R_{t+1} = \frac{P_{t+1} - P_t}{P_t}$$

- where P_t Price at time t
- Example:
  - Buy stock at $P_t = \$100$
  - Sell stock at $P_{t+1} = \$110$
  - Return is $R_{t+1} = \frac{P_{t+1} - P_t}{P_t} = \frac{\$110 - \$100}{\$100} = 10\%$

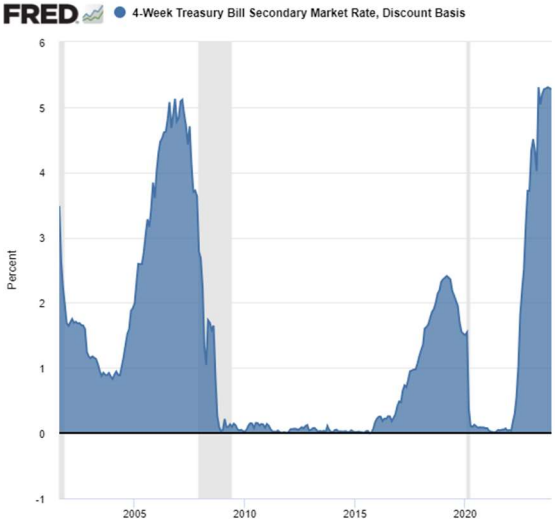
Decompose Returns

$$R = R_f + RP$$

- Risk-free rate R_f
  - The risk-free rate reflects the time value of money
  - How willing an investor is to give up $1 today for $$$ in the future
  - The more impatient a consumer, the more future compensation he will demand to delay consumption and hence the higher the risk-free rate
- Risk premium RP
  - Investors who dislike risk are risk averse and demand risk premiums for bearing risk
  - The risk-premium increases when:
    - The riskiness of an asset increases
    - Investors become more risk averse

Risk-free Rate R_f

- Treasury bills have no risk of default, very short-term maturities, a known return and are traded in active markets
- They are the closest approximation available to a riskless investment
- 30-day Treasury bill rate will be used to approximate the monthly riskless rate of interest



Empirical Risk Premiums

- Measure risk premiums in the data by looking at the difference between an asset's expected return and the risk-free rate

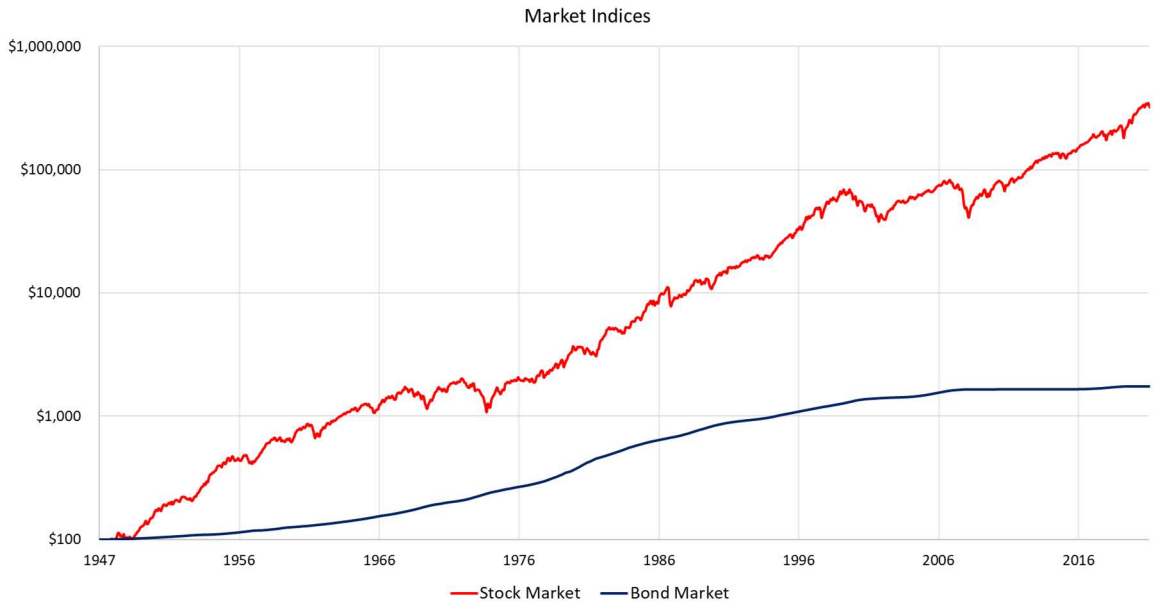
$$RP = E[R] - R^f$$

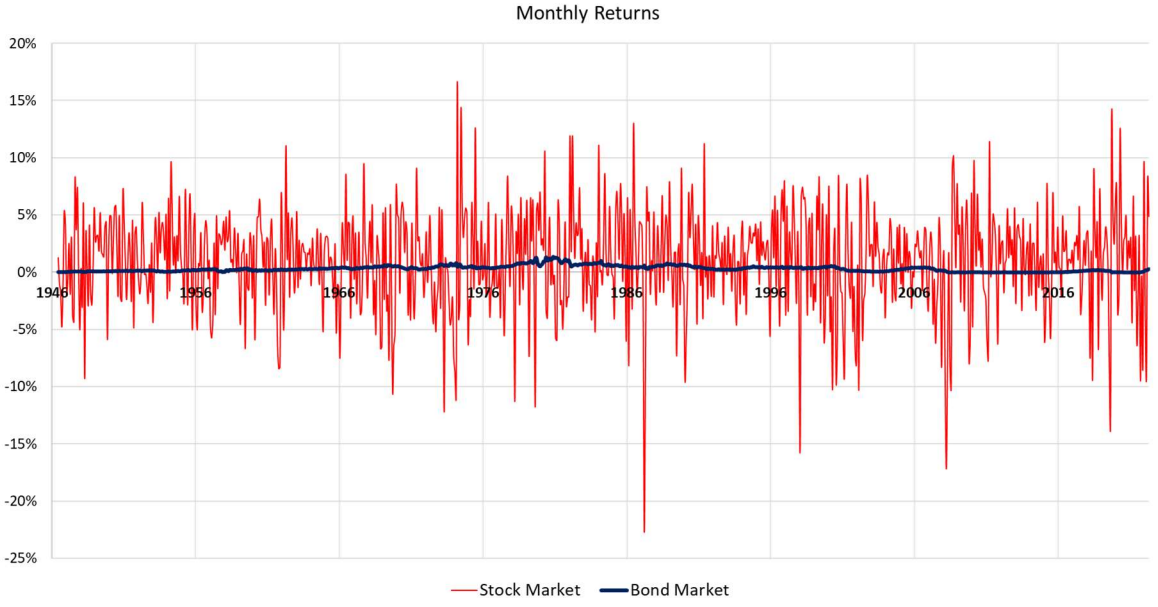
- We seek to understand what are the important risk factors that determine risk premiums
- Bad stocks have high returns
  - High returns reflect compensation for time and risk

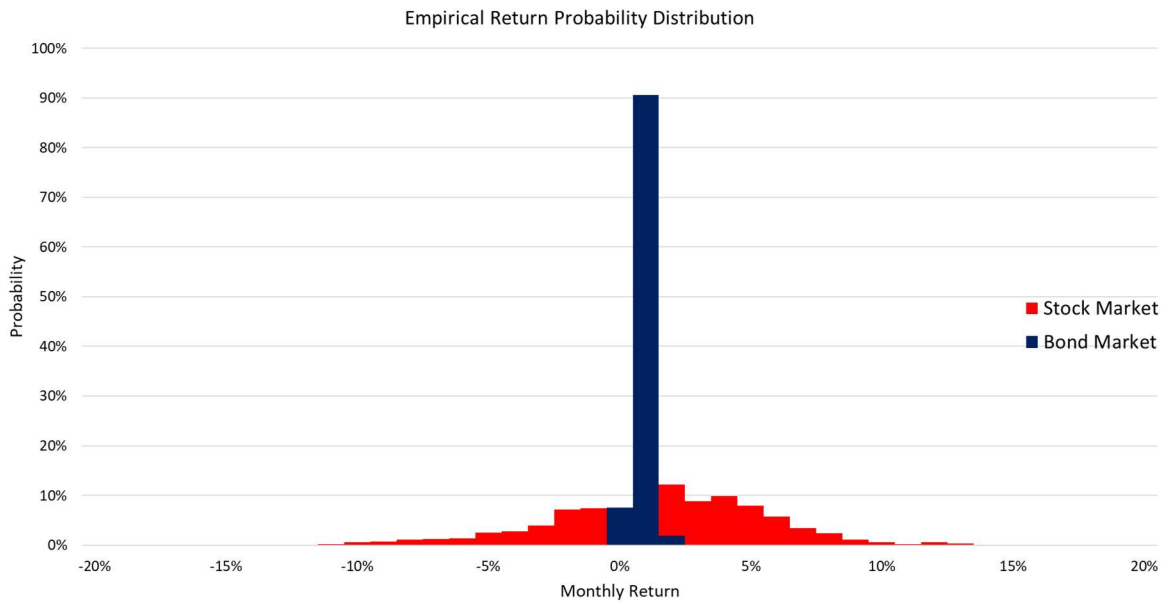
Sharpe Ratio: Reward-to-Risk

- Excess Return: $EXR_t = R_t - R_f$
  - The difference between an asset's return and the risk-free rate at time t
- Risk Premium: $RP = E[EXR_t] = E[R_t] - R_f$
  - The expected (i.e. average) excess return
- Sharpe Ratio
  - The ratio of an asset's risk premium to standard deviation
  - Represents reward-risk tradeoff

$$Sharpe\ Ratio = \frac{E[R_t] - R_f}{\sigma} = \frac{RP}{\sigma}$$


Market Indices
$1,000,000
$100,000
$10,000
$1,000
$100
1947
1956
1966
1976
1986
1996
2006
2016
Stock Market
Bond Market


Monthly Returns
20%
15%
10%
5%
0%
-5%
-10%
-15%
-20%
-25%
1946
1956
1966
1976
1986
1996
2006
2016
Stock Market
Bond Market



Implications

Stat	Finance	Observe	Implication
Mean	Reward	12%	Double wealth ~6 yr
Standard Deviation	Risk	15%	95% of years have returns [-22%,37%]
Sharpe	Reward-Risk Ratio	0.50	Benchmark to grade other investments

Portfolio Theory

- Given a menu of assets, what do we invest in and how much?
  1. Asset allocation: what assets (e.g. bonds vs. stocks)
  2. Security selection: what securities (e.g. Apple vs. Google)
- How do we define best? Sharpe ratio
- Choose investment weights w_i to maximize portfolio Sharpe (Reward-to-Risk) ratio:

$$\max \frac{E[R^P] - R^f}{\sigma_P}$$

Portfolio Theory

- R_{it} is the return on asset i at time t
- w_i weight in asset i
- Portfolio return at time t

$$R_t^P = w_1 \cdot R_{1t} + \cdots + w_n \cdot R_{nt}$$

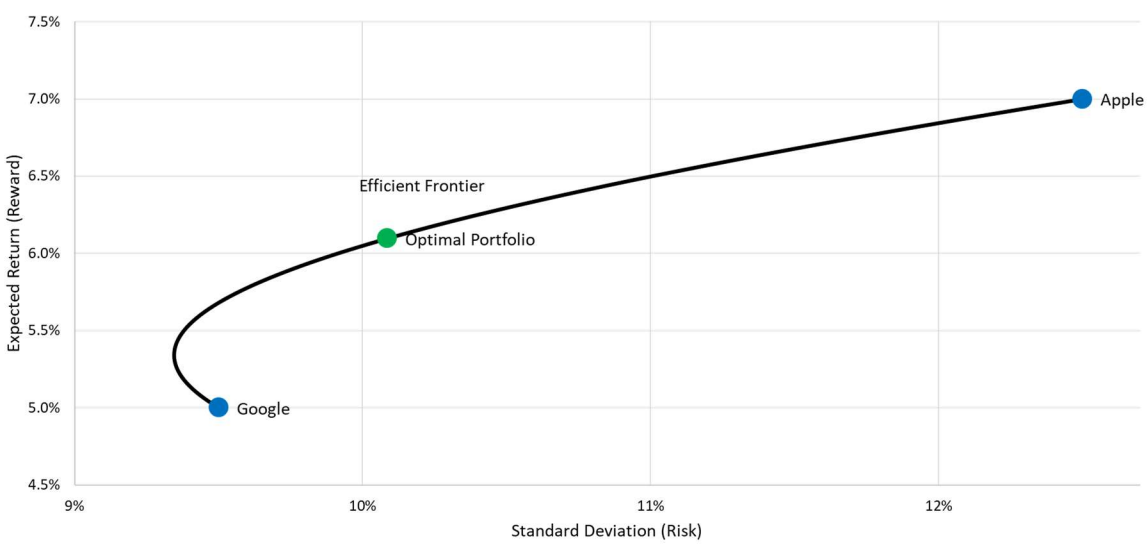
Portfolio Statistics

Given historical data

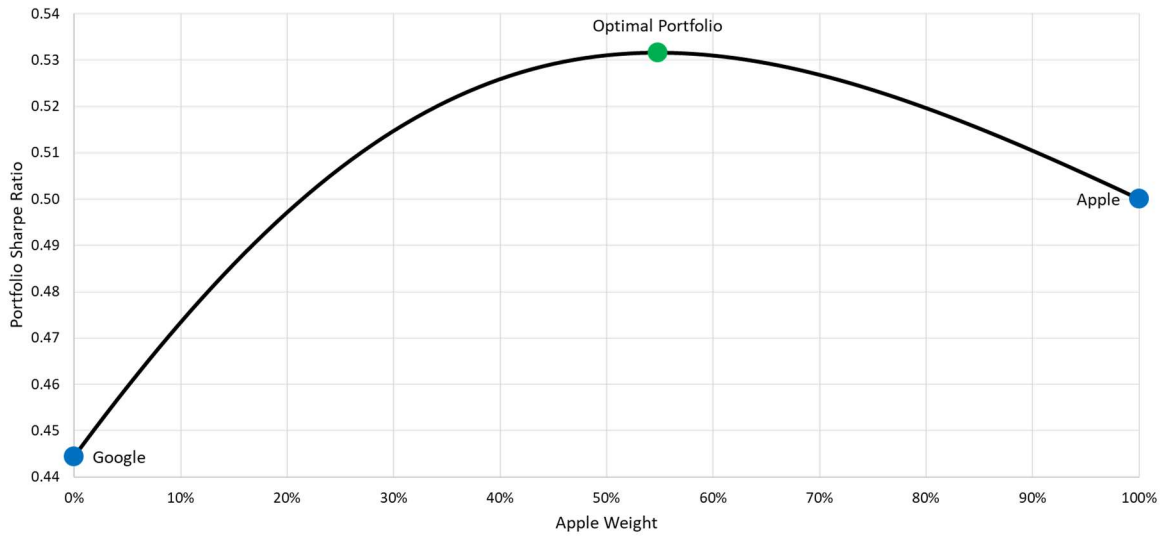
- $E[R^P] = \text{average}\left(R_t^P\right)$
- $\sigma_P = \text{stdev.s}\left(R_t^P\right)$

- Analytical formulas for two assets
  - $E[R^P] = w_1 E[R_1] + w_2 E[R_2]$
  - $\sigma_P = \sqrt{w_1^2\sigma_1^2 + w_2^2\sigma_2^2 + 2w_1w_2\sigma_{12}}$

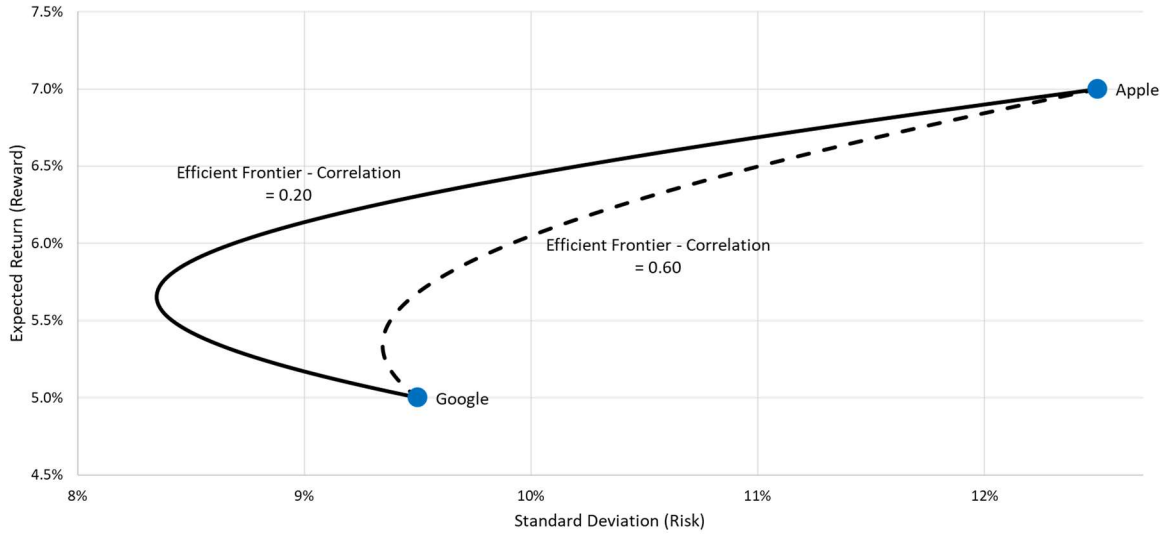
Security Selection



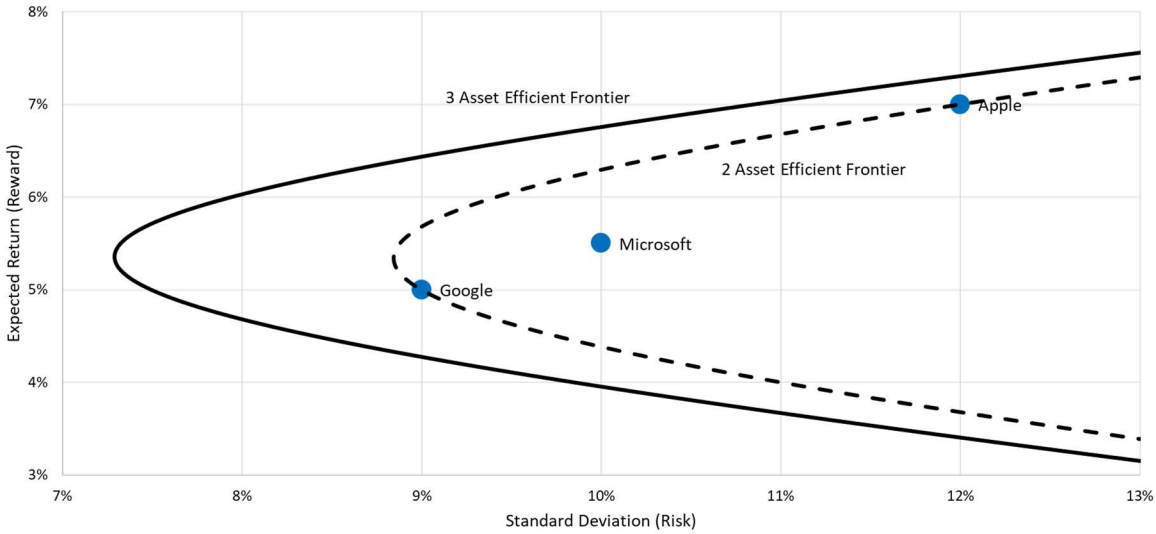
Security Selection



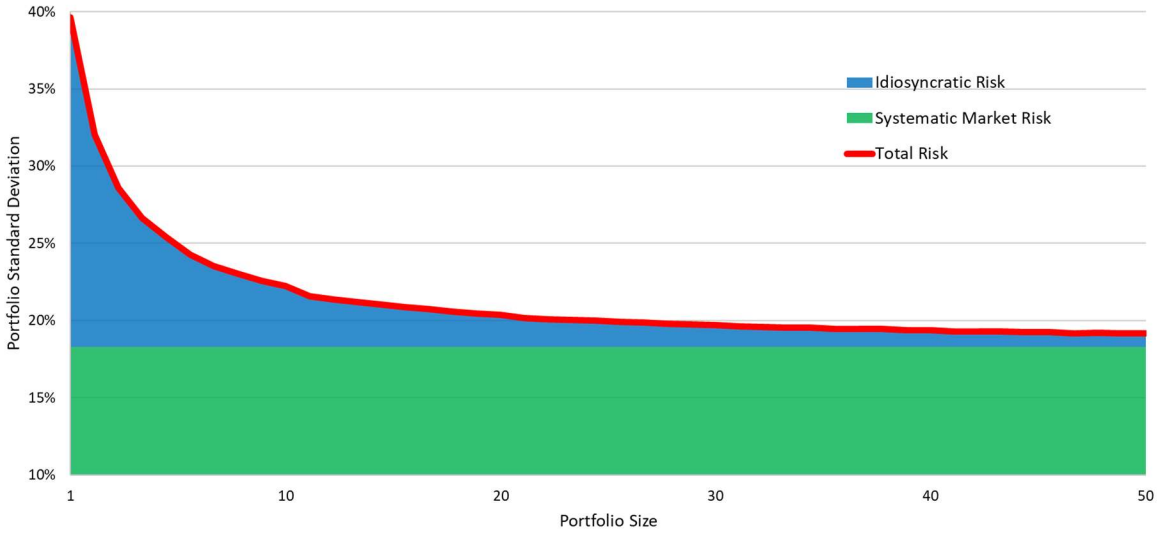
Diversification & Correlation



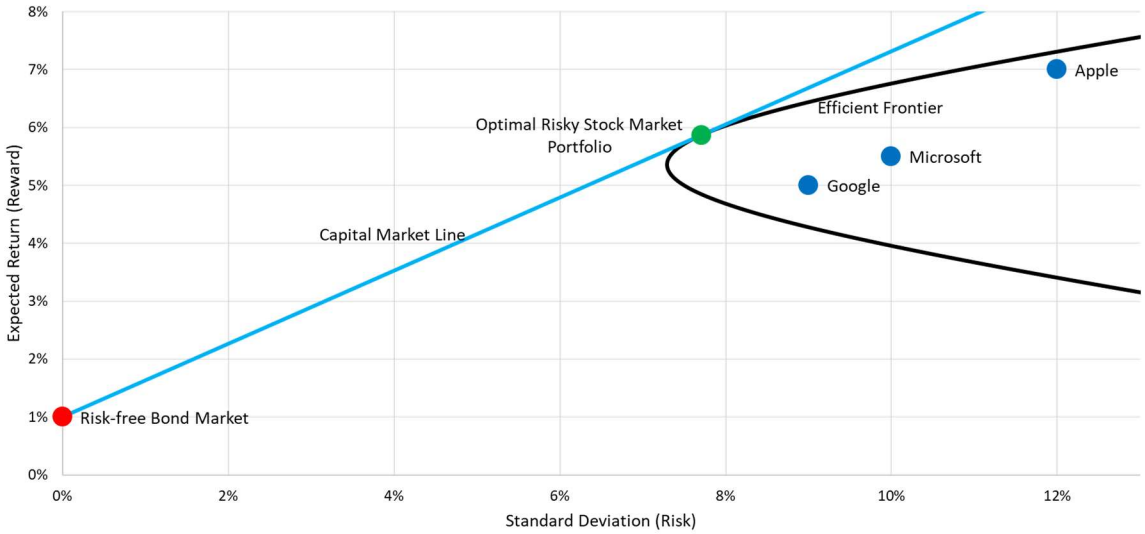
Diversification & Number of Assets



Diversification & Optimal Number of Assets



Portfolio Optimization



Portfolio Optimization

Summary

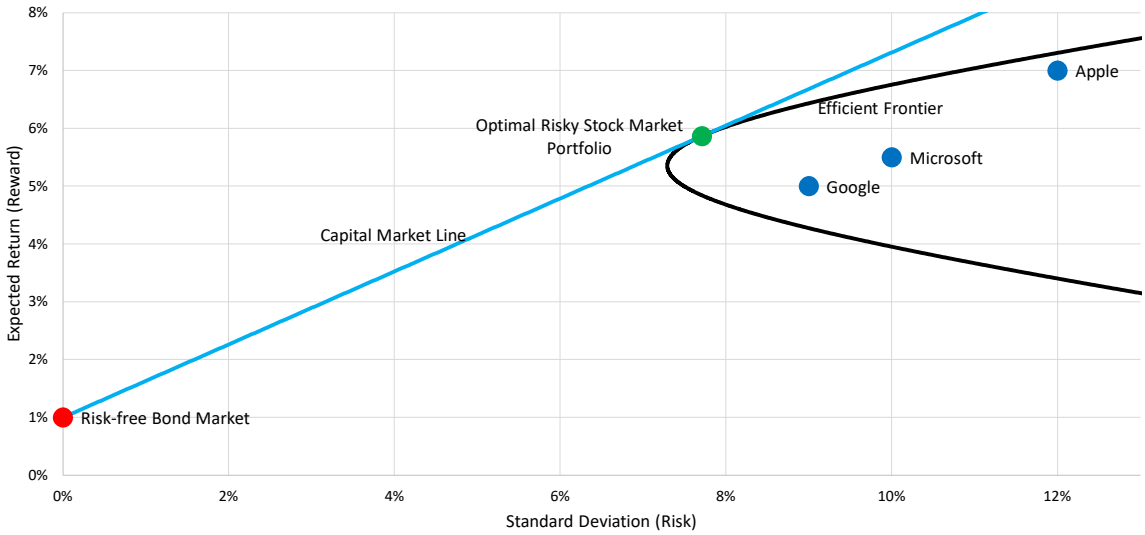
- Security selection
  - Calculate optimal risky market portfolio
  - Same for all investors
  - Only a function of assets' riskiness & correlation structure
- Asset allocation
  - Choose personal portfolio as a weighted average of risky market portfolio and risk-free bond market
  - Unique to each investors' risk preference

Portfolio Optimization

Summary

- Diversification allows us to obtain portfolios with higher Sharpe ratios relative to the portfolio's holdings.
- When assets are not perfectly correlated, the riskiness of the portfolio falls, resulting in a higher Sharpe ratio.
- Intuition: If two assets are negatively correlated, when one does poorly, the other does well on average. The assets hedge each other's risk, thus decreasing the overall risk of the portfolio.

Capital Market Line



Capital Market Line

How do we price an asset on the CML?

- Plot R_i vs. σ_i
- y-intercept $= R_f$
- Slope $= \frac{R_M - R_f}{\sigma_M}$
  - Sharpe ratio of Market

$$R_i = R_f + \left(\frac{R_M - R_f}{\sigma_M}\right)\sigma_i$$

CAPM Derivation

How do we price an asset not on the CML?

$$\begin{aligned} R_i &= R_f + \left(\frac{R_M - R_f}{\sigma_M}\right) \cdot \sigma_i \cdot \rho_{iM} \\ &= R_f + \left(\frac{R_M - R_f}{\sigma_M}\right) \cdot \sigma_i \cdot \frac{\sigma_{iM}}{\sigma_i \cdot \sigma_M} \\ &= R_f + \frac{\sigma_{iM}}{\sigma_M^2}\left(R_M - R_f\right) \\ &= R_f + \beta_i\left(R_M - R_f\right) \end{aligned}$$

- σ_i and σ_M are the standard deviation of asset i and the market
- $\rho_{iM} = \frac{\sigma_{iM}}{\sigma_i \sigma_M}$: Correlation between asset i and market
- σ_{iM}: Covariance between asset i and market
- $\beta_i = \frac{\sigma_{iM}}{\sigma_M^2}$: Asset i's beta

CAPM

$$R_i = R_f + \beta_i(R_M - R_f)$$

- R_i: Expected return on asset i
- β_i: Asset i's beta
- R_M: Expected return on market
- R_f: Risk-free rate

CAPM Interpretation

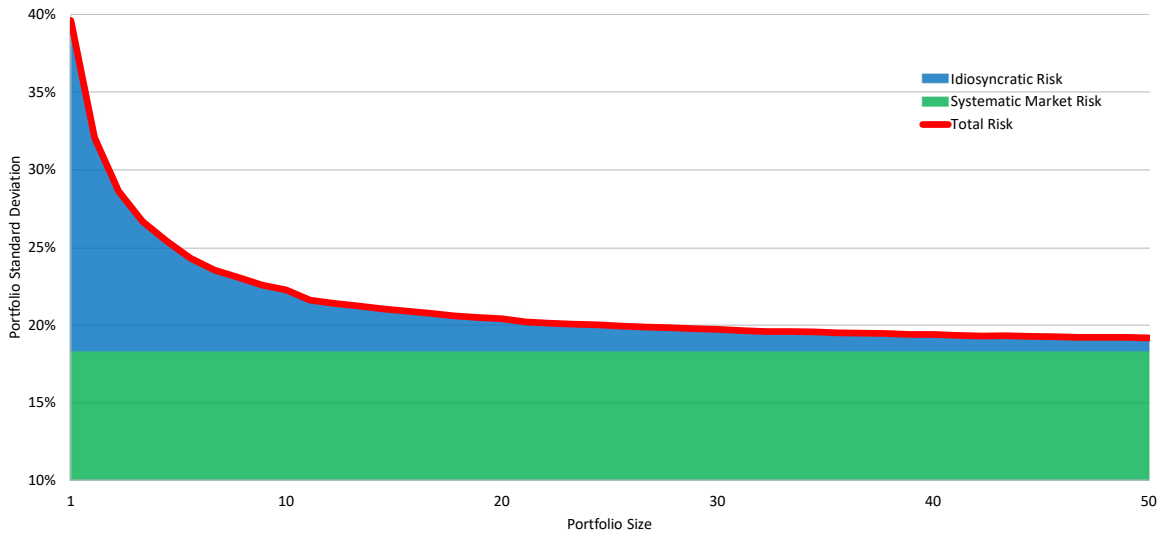
$$R_i - R_f = \beta_i \times (R_M - R_f)$$

- $R_i - R_f$: Risk premium on asset i
  - The price of asset i's risk
- β_i: Comovement of asset i with the market
  - Quantity of market risk
- $R_M - R_f$: Market risk premium
  - Price of market risk

- Interpretation

$$\text{Price of Asset } i'\text{s Risk} = \text{Quantity of Market Risk} \times \text{Price of Market Risk}$$

Diversification & Risk Compensation



Diversification & Risk Compensation

- An asset's total risk can be decomposed into two sources

$$\text{Total Risk} = \text{Systematic Risk} + \text{Idiosyncratic Risk}$$

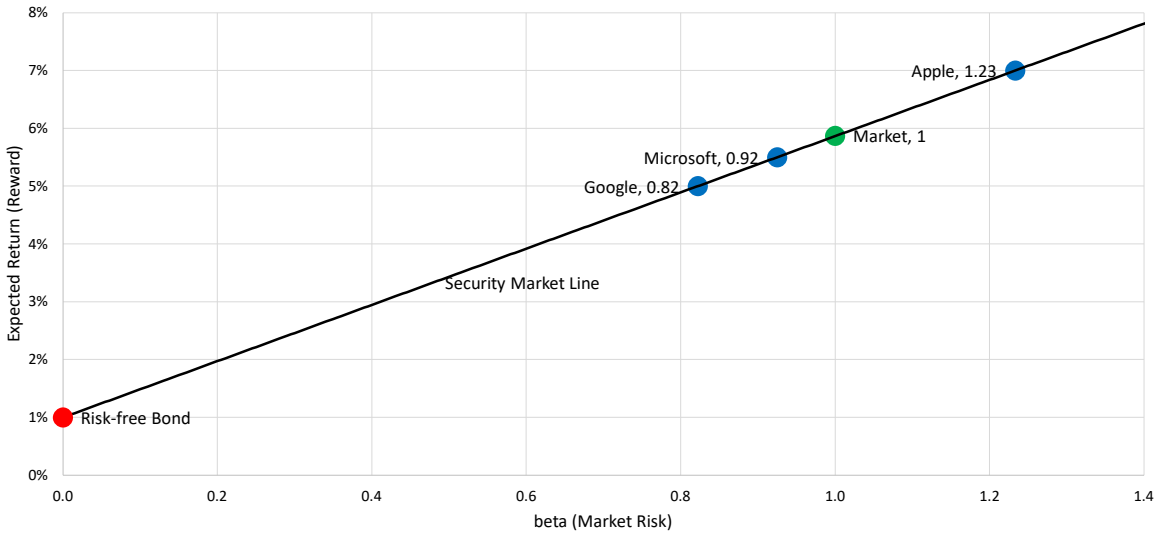
- The CAPM says that only systematic risk is priced, i.e. investors only receive a risk premium for an asset's exposure to market risk, measured by β

Security Market Line

- Plot R_i vs. β_i
- y-intercept $= R_f$
- Slope $= R_M - R_f$
  - Market risk premium

$$R_i = R_f + \beta_i\left(R_M - R_f\right)$$

Security Market Line



Estimating CAPM via Regression Analysis

- CAPM: $R_i - R_f = \beta_i \times \left(R_M - R_f\right)$
- Regression: $y = \alpha + \beta \cdot x$
  - Seek to explain y with x
  - Regress y onto x
  - Estimate y-intercept α and slope β
- Apply regression to CAPM
  - $y \equiv R_i - R_f$
  - $x \equiv R_M - R_f$
  - Seek to explain asset i's excess returns with market excess returns
  - Regress asset i's excess returns onto market excess returns

Estimating CAPM via Regression Analysis

- Regression

$$\hat{y} = \hat{\alpha} + \hat{\beta} \cdot x$$

  - Given estimated coefficients $\hat{\alpha}$ and $\hat{\beta}$, model predicts $\hat{y}$ if x occurs
- Mathematically, we have

$$\bar{y} = \hat{\alpha} + \hat{\beta} \cdot \bar{x}$$

  - $\bar{y}$ and $\bar{x}$ are average of y and x
- CAPM

$$E[R_i] - R_f = \hat{\alpha}_i + \hat{\beta}_i \cdot \left(E[R_M] - R_f\right)$$

$$RP_i = \hat{\alpha}_i + \hat{\beta}_i \cdot RP_M$$

  - The average excess return is the risk premium: $RP = E[R] - R_f$

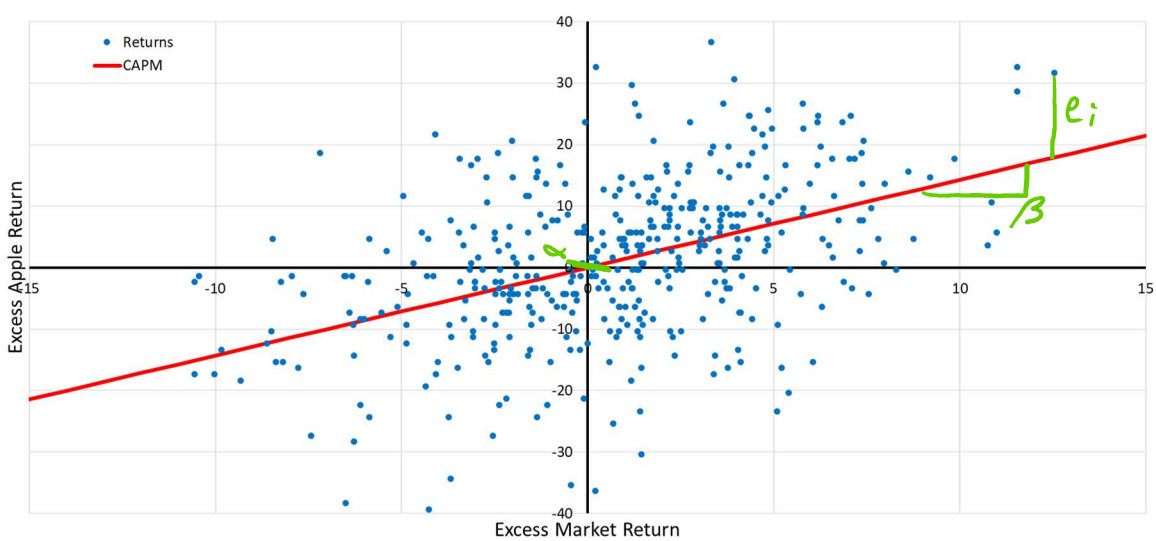
Risk Decomposition

- To price asset i, we can decompose asset i's risk premium into
$$RP_i = \hat{\alpha}_i + \hat{\beta}_i \cdot RP_M$$
- RP_i: Risk premium on asset i
  - The price of asset i's risk
- β_i: Comovement of asset i with the market
  - Quantity of market risk
- RP_M: Market risk premium
  - Price of market risk

- Interpretation

$$\text{Price of Asset } i\text{'s Risk} = \text{Quantity of Market Risk} \times \text{Price of Market Risk}$$

Apple's CAPM Estimation



Apple's CAPM Interpretation

- Market's monthly risk premium: $R_M - R_f = 0.62\%$
- Apple's monthly risk premium: $R_A - R_f = 1.84\%$
- Apple's beta: $\beta_A = 1.42$

- Risk premium decomposition:

$$R_A - R_f = \beta_A \times \left(R_M - R_f\right)$$
$$1.84\% \approx 1.42 \times 0.62\%$$

- Annualized:

$$22\% \approx 1.42 \times 7.4\%$$

CAPM Estimation Interpretations

- Estimate α, β

$$R_i - R_f = \alpha + \beta \cdot (R_M - R_F)$$

Important estimates to consider:

- Finance theory says $\alpha = 0$. If it is not, how do we interpret this?
- If $\beta = 0$, market risk factor is unimportant
- If $\beta > 1$, asset is riskier than market and commands higher risk premium

Inference

Is the CAPM model good statistically?

1. Hypothesis Test: $\beta = 0$
   - $\beta \neq 0 \Rightarrow$ Market risk factor is important
2. ANOVA: r^2
   - $r^2 \ll 100\% \Rightarrow$ Market is not the only important factor

Finance Theory

$\alpha = 0$

- If $\alpha > 0$, two interpretations
  1. Skill
  2. Missing risk factors
  - Empirically we cannot find persistent $\alpha > 0$

$\beta > 1$

- Asset is riskier than market and commands higher risk premium

Corporate Bonds & Cost of Debt R_D

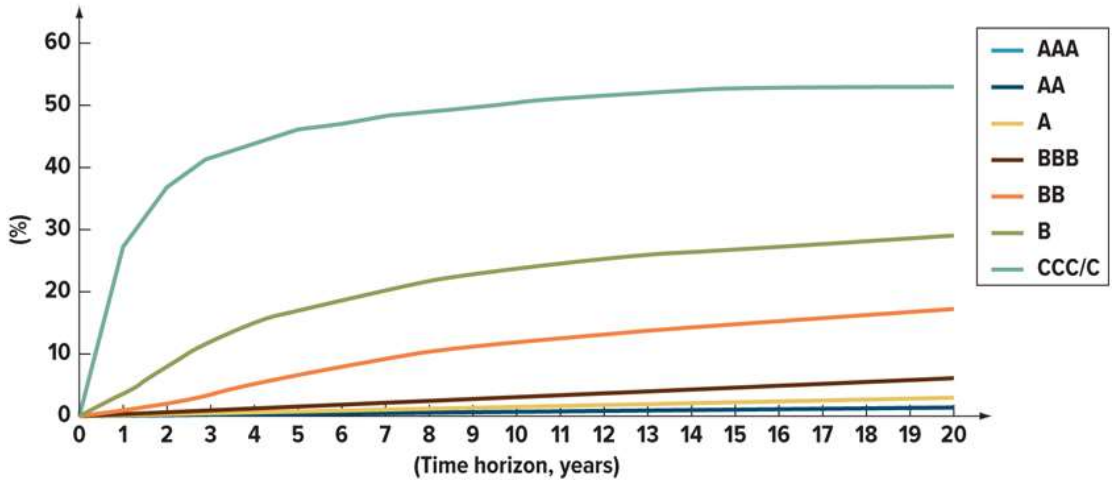
$$R_D = R_f + Default\ Spread$$

Bond Ratings

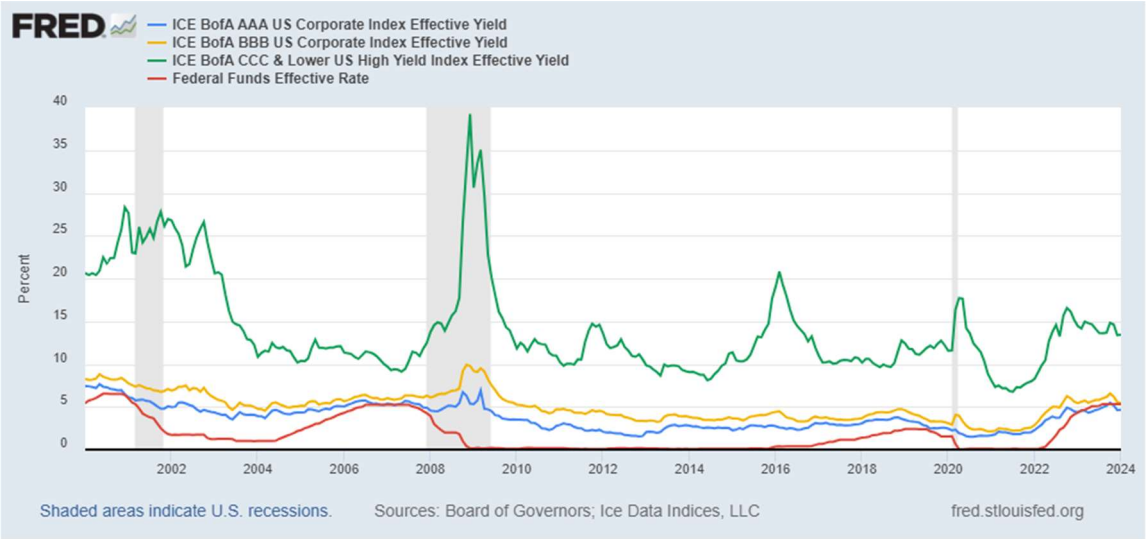
	Investment-Quality Bond Ratings				Low-Quality, Speculative, and/or "Junk" Bond Ratings					
	High Grade		Medium Grade		Low Grade		Very Low Grade			
Standard & Poor's	AAA	AA	A	BBB	BB	B	CCC	CC	C	D
Moody's	Aaa	Aa	A	Baa	Ba	B	Caa	Ca	C	

Moody's	S&P	
Aaa	AAA	Debt rated Aaa and AAA has the highest rating. Capacity to pay interest and principal is extremely strong.
Aa	AA	Debt rated Aa and AA has a very strong capacity to pay interest and repay principal. Together with the highest rating, this group comprises the high-grade bond class.
A	A	Debt rated A has a strong capacity to pay interest and repay principal, although it is somewhat more susceptible to the adverse effects of changes in circumstances and economic conditions than debt in higher-rated categories.
Baa	BBB	Debt rated Baa and BBB is regarded as having an adequate capacity to pay interest and repay principal. Whereas it normally exhibits adequate protection parameters, adverse economic conditions or changing circumstances are more likely to lead to a weakened capacity to pay interest and repay principal for debt in this category than in higher-rated categories. These bonds are medium-grade obligations.
Ba; B Caa Ca C	BB; B CCC CC C	Debt rated in these categories is regarded, on balance, as predominantly speculative with respect to capacity to pay interest and repay principal in accordance with the terms of the obligation. BB and Ba indicate the lowest degree of speculation, and Ca, CC, and C the highest degree of speculation. Although such debt is likely to have some quality and protective characteristics, these are outweighed by large uncertainties or major risk exposures to adverse conditions. Issues rated C by Moody's are typically in default.
	D	Debt rated D is in default, and payment of interest and/or repayment of principal is in arrears.

Bond Ratings & Default Rate



Credit Spreads over Time



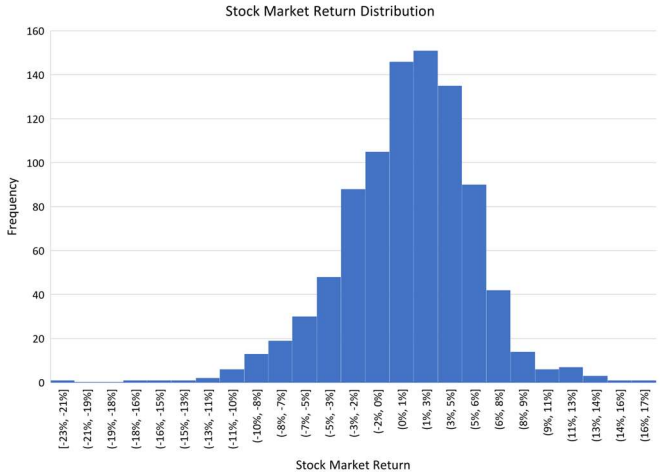
Risk & Return Analytics Primer

Professor Philip Howard
howardpd@wfu.edu

WAKE FOREST UNIVERSITY
SCHOOL of BUSINESS

Descriptive Statistics

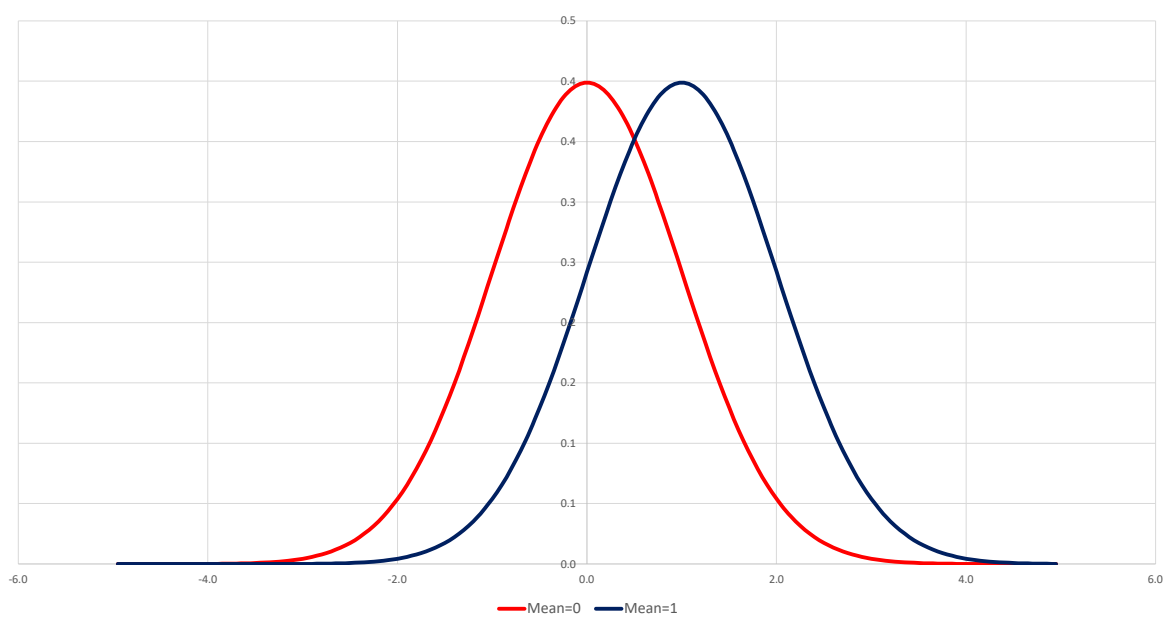
- Summarize a distribution's shape



Measure of Center: Mean

- Excel: =average(data)
- Also denoted as μ or $\bar{x}$
- Mean, average, expected value all synonymous

$$E[x] = \frac{1}{n}\sum_{i=1}^{n} x_i$$



Measures of Dispersion

- Dispersion refers to the degree of variation in the data, i.e. the spread of the distribution
- Important measures:
  - Variance σ^2
  - Standard deviation σ

Measures of Dispersion: Variance

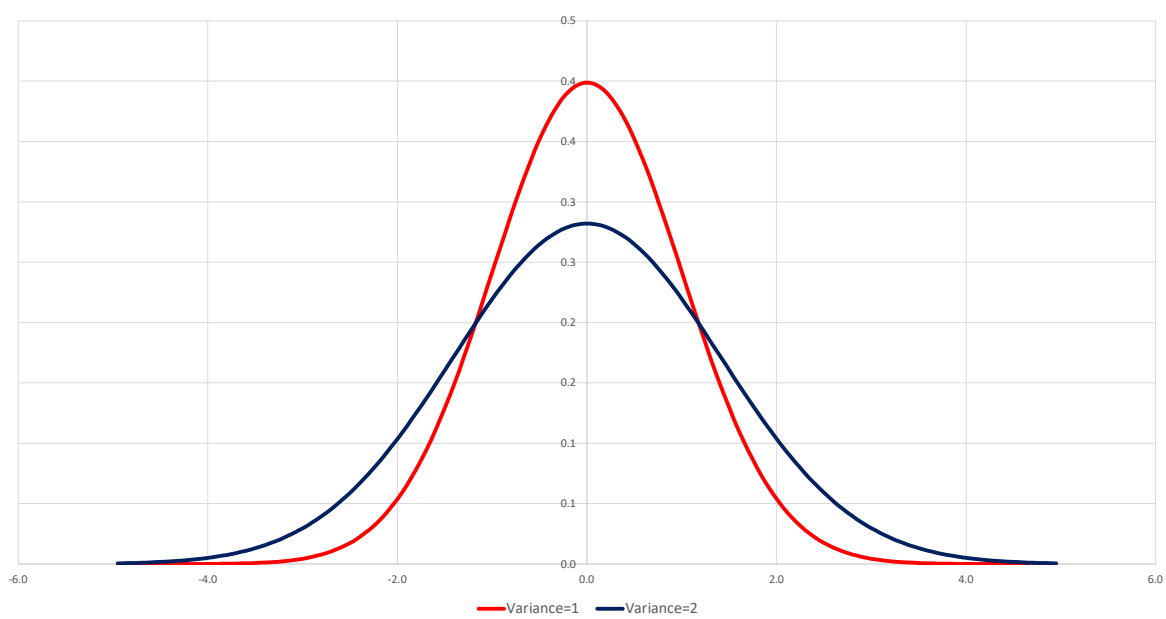
- Excel: =var.s(data)

$$Var(x) = \sigma_x^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2$$

Measures of Dispersion: Standard Deviation

- The units of the variance is the square of the dimension of the observations, whereas the units of the standard deviation is the same as the data. This makes the standard deviation more practical to use in applications.
- $\sigma \geq 0$
- Excel: =stdev.s(data)

$$SD(X) = \sigma_x = \sqrt{\sigma_x^2}$$



Measures of Association

- Two variables have a strong statistical relationship with one another if they appear to move together
- When two variables appear to be related, you might suspect a cause-and-effect relationship
- Sometimes, however, statistical relationships exist even though a change in one variable is not caused by a change in the other. Thus, correlation is not causation.

Measures of Association: Covariance

- Covariance is a measure of the linear association between two variables
- The covariance between X and Y is the average of the product of the deviations of each pair of observations from their respective means.
- Excel function: =covariance.s(data1,data2)

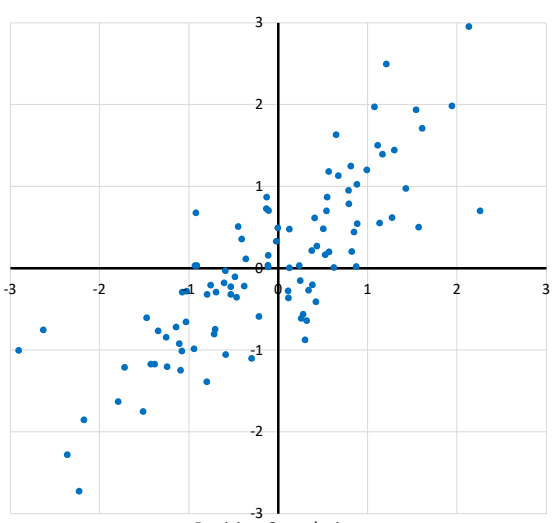
$$Cov(X,Y) = \sigma_{xy} = \frac{1}{n-1}\sum(x_i - \bar{x})(y_i - \bar{y})$$

Measures of Association: Correlation

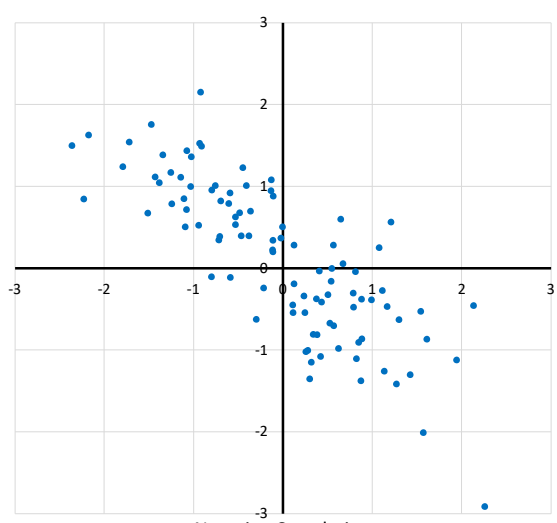
- Correlation is a measure of the linear relationship between two variables, which does not depend on the units of measurement
- The correlation coefficient is between -1 and 1
  - $r > 0$ indicates a positive relationship
    - When X goes up, Y goes up on average
  - $r < 0$ indicates a negative relationship
    - When X goes up, Y goes down on average
  - The larger $|r|$, the stronger the linear relationship
- Correlation does not imply causality
- Excel function: =correl(data1,data2)
- Also denoted r_{xy}

$$\rho_{xy} = \frac{Cov(X,Y)}{SD(X) \cdot SD(Y)} = \frac{\sigma_{xy}}{\sigma_x \sigma_y}$$

Examples of Correlation

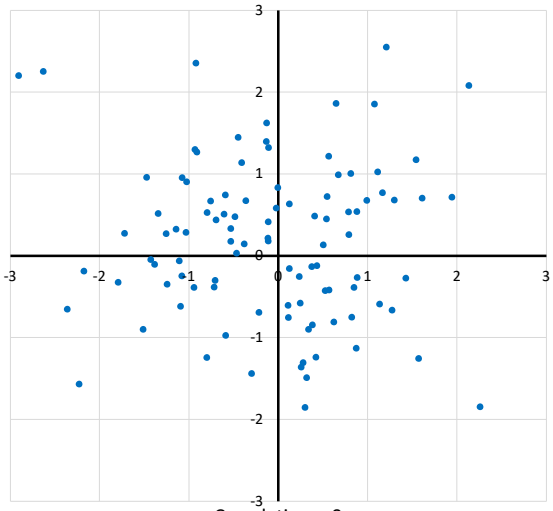


Positive Correlation

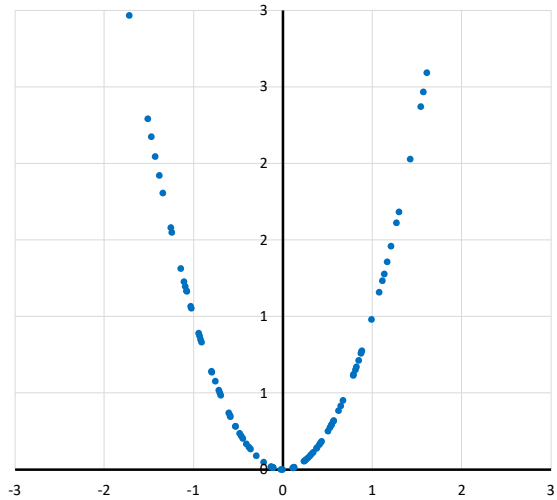


Negative Correlation

Examples of Correlation



Correlation = 0



Correlation = 0

Random Variables

- Random variable has random outcomes
- Outcomes have probabilities
- Two types of probability distributions
  - Discrete: outcomes are discrete values
  - Continuous: outcomes over continuous interval
  - Probability distributions sum to 1

Linear Combinations of Random Variables

Let a, b and c be constants and X, Y and Z be random variables

- $E[aX + bY] = aE[X] + bE[Y]$
- $Var(aX + bY) = a^2Var(X) + b^2Var(Y) + 2ab \cdot Cov(X, Y)$
- $Cov(aX + bY, cZ) = ac \cdot Cov(X, Z) + bc \cdot Cov(Y, Z)$

- Note: $Cov(X, X) \equiv Var(X)$

Numerical Optimization

Objective function
- Value we seek to minimize or maximize

Decision variables
- Variables we can change

Constraints
- Constraints on model

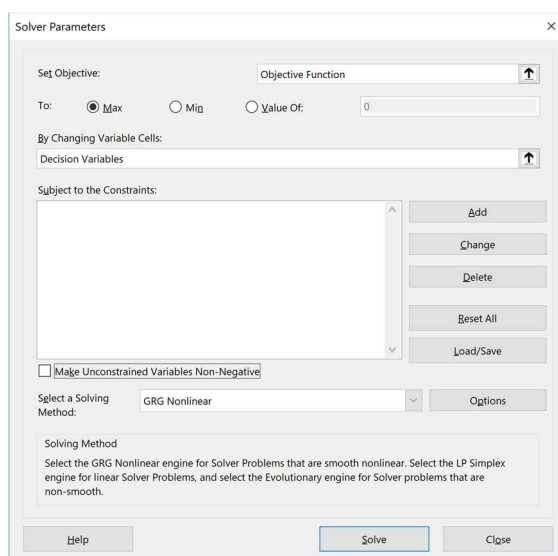
Numerical Optimization

$$\max_{\text{Decision Variables}} \quad \text{Objective Function}$$

$$\text{s.t.} \quad \text{Constraints}$$

Solving Optimization Models

Solver is an Excel add-in for solving optimization problems



Application: Portfolio Theory

- Given a menu of assets, what do we invest in and how much?
- n assets
  - R_{it} is the return on asset i at time t
- Choose investment weights
  - w_i weight in asset i
- Given investment weights, portfolio return at time t is

$$R_t^P = w_1 \cdot R_{1t} + \cdots + w_n \cdot R_{nt}$$

Objective Function

Maximize Sharpe ratio

- Reward-to-Risk

$$\max \frac{E[R^P] - R^f}{\sigma_P}$$

Decision Variables

Choose investment weights

- w_i weight in asset i

Constraints

- Constraints must be expressed as inequalities or equations
- Look for key words in word statements of constraints

cannot exceed	$\leq$
at least	$\geq$
exactly	=

Constraints

Sum of weights totals 100%

$$\sum_{i=1}^{n} w_i = 1$$

No short selling

$$w_i \geq 0$$

Solutions

Unbounded Solution

- Objective can be increased without bound to infinity for a maximization problem or decreased to negative infinity for a minimization problem

Infeasible Solution

- Solution that violates at least one constraint
- Model is infeasible if all solutions are infeasible

Feasible Solution

- Any solution that satisfies all of the constraints

Optimal Solution

- Feasible solution achieves best objective function

Unique Optimal Solution

- Exactly one optimal solution

Constraints

- Binding constraint
  - Constraint holds with equality at optimal solution
- Slack constraint
  - Constraint is not binding

Sensitivity Analysis

What happens when we change the:

- Objective function?
- Constraints?
- Parameters?
- How sensitive are our optimal decision variables?

- Portfolio theory we will explore changing:
  - Objective function: alternative risk-reward preferences
  - Constraints: investment limits
  - Parameters: correlation, number of assets

Regression Analysis

$$y = \alpha + \beta_1 \cdot x_1 + \cdots + \beta_n \cdot x_n + \varepsilon$$

- y: Dependent variable
- x_i: Independent variables
- α: y-intercept
- β_i: Slope coefficients
- ε: Unknown, true population errors

Regression Analysis

$$y = \alpha + \beta_1 \cdot x_1 + \cdots + \beta_n \cdot x_n + \varepsilon$$

- We seek to explain y with x_i's
- "Regress y onto x"
- Observe a sample of y and x
- Estimate α and β_i
  - Estimates denoted $\hat{\alpha}$ and $\hat{\beta}_i$
- Do not observe ε
  - Assume $\varepsilon \sim N(0, \sigma^2)$
  - σ^2 unknown and needs to be estimated

Model Estimation & Prediction

Estimate the model

$$\hat{y} = \hat{\alpha} + \hat{\beta}_1 \cdot x_1 + \cdots + \hat{\beta}_n \cdot x_n$$

- $\hat{\alpha}$, $\hat{\beta}_i$ are estimated
- $\hat{y}$ is the predicted value given x and estimated $\hat{\alpha}$ and $\hat{\beta}_i$

Estimation

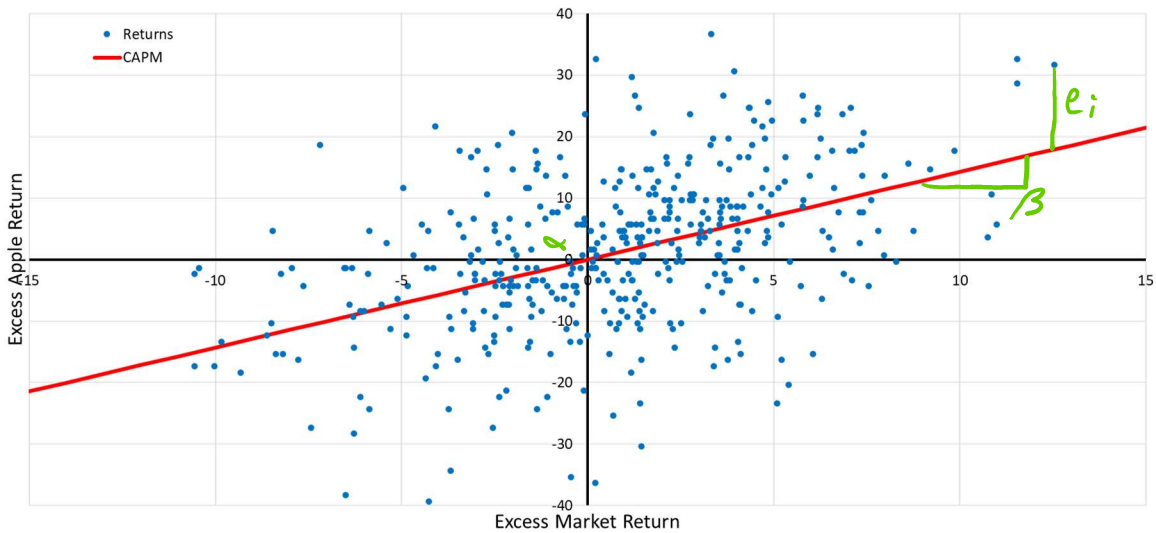
- Goal is to choose $\hat{\alpha}$, $\hat{\beta}$ to find the "best" line
- How do we define best? Minimize the sum of the squared errors

$$\min_{\{\alpha,\beta\}} \sum_{i=1}^{n} e_i^2$$

CAPM analytical solution:

- $\hat{\beta} = \frac{\sigma_{iM}}{\sigma_M^2}$

Apple's Estimated CAPM



Regression Application: CAPM

- Mathematically, we have

$$\bar{y} = \hat{\alpha} + \hat{\beta} \cdot \bar{x}$$

- CAPM Interpretation

$$RP_i = \hat{\alpha} + \hat{\beta} \cdot RP_M$$

Coefficient of Determination

$$R^2 = \frac{SSR}{SST}$$

- Percent variation of y explained by x
- $0 \leq R^2 \leq 1$
- Closer to 1, the better the model

Coefficient of Determination & Correlation

For models with only a single x

- Correlation $R = \frac{Cov(x,y)}{SD(x) \cdot SD(y)} = \frac{\sigma_{xy}}{\sigma_x \sigma_y}$
- Can calculate the coefficient of determination from two perspectives
  - ANOVA: $R^2 = \frac{SSR}{SST}$
  - Correlation: $R^2 = (R)^2$

Hypothesis Tests: Model Estimates

- When we estimate α and β, we want to know how confident we are in our estimates
- Statistical and economically motivated tests
- When we perform a hypothesis test, we will always compare two hypothesis: a null and an alternative
- Goal is to see which hypothesis is supported by data statistically

Hypothesis Tests

- Null hypothesis:

$$H_0: \beta = \beta_0$$

- Alternative hypothesis:

$$H_a: \beta \neq \beta_0$$

Hypothesis Tests

- P-value
  - "Probability of estimating $\hat{\beta}$ given the null is true"

- Significance level α
  - Probability of incorrectly rejecting null
  - Typical levels of 1%, 5%, or 10% in practice
  - If you want to be more confident of result, use smaller α
    - Confidence $= (1 - \alpha)100\%$
    - Smaller α is a tougher hurdle to discover a result

- Test conclusion
  - Reject null if P-value $<$ significance level

Significance Level

Note we have two alphas:

- Significance level α
  - From statistics
- y-intercept α
  - From finance, called "Jensen's α"
- Different, unrelated values

Inference

Is the CAPM model good statistically?

1. Hypothesis Test: $\beta = 0$
   - $\beta \neq 0 \Rightarrow$ Market risk factor is important
2. ANOVA: r^2
   - $r^2 \ll 100\% \Rightarrow$ Market is not the only important factor

Finance Theory

$\alpha = 0$

- If $\alpha > 0$, two interpretations
  1. Skill
  2. Missing risk factors
  - Empirically we cannot find persistent $\alpha > 0$

$\beta > 1$

- Asset is riskier than market and commands higher risk premium